

Siddhant Arora

Software Developer | Competitive Mathematician | Open-Source Builder
Calgary, AB, Canada • siddaroraleo@gmail.com • <https://github.com/siddhantarora7>

SUMMARY

Competitive programmer and mathematician focused on algorithms, systems, and open-source education tools. USACO Gold competitor, and CCC Top 3% nationally. Currently building open-source platforms for math olympiad training. Passionate about solving difficult problems and building tools that expand access to high-quality technical education.

EDUCATION

Westmount Charter School (GPA: 96/100)

Sophomore (Grade 10) • Calgary, AB • 2025 – Present

- One of Calgary's most academically selective charter schools, specializing in gifted education.
- Consistently maintained Honor Roll standing; active contributor to the school's STEM culture.

COMPETITIONS & CERTIFICATIONS

Canadian Computing Competition (CCC)

Honor Roll • Top 3% Nationally • 2026

- Achieved Honor Roll distinction, placing in the top 3% of all Canadian competitors.

USA Computing Olympiad (USACO) — Gold Division

2024 – 2025

- Reached Gold Division, the second-highest tier, through self-directed study of advanced algorithms and data structures.

CalgaryHacks — Tier 2, 3rd Place

City-Wide Hackathon • 500+ Participants • 2026

- Placed 3rd in Tier 2 at Calgary's largest annual student hackathon (500+ participants), building SpinFilter — a media bias detection platform — in 24 hours.

CS50 — Harvard / edX

Certificate of Completion • 2024

- Completed Harvard's rigorous introduction to computer science covering C, Python, SQL, and web development.

PROJECTS

usamo.guide — COO & SWE

usamo.guide • Open-Source Platform • In Active Development • 2025 – Present

- Leading development of a free, centralized math olympiad preparation platform — structured across four tiers from AMC 8 foundations to USAMO with 300+ curated problems across 20 modules.
- Architecting full-stack infrastructure: tiered problem database, structured learning paths, user progress tracking, authentication, and contributor pipelines.
- Managing a cross-functional team of 30+ people across roadmap planning, code reviews, and deployment.
- Employed SEO techniques and marketed to attain 50k+ site visits and 5k+ registered users in under a month..
- Stack: Python, Flask, ReactJS, JavaScript, SQL, HTML/CSS.

SpinFilter — CalgaryHacks 2026 — Tier 2, 3rd Place

CalgaryHacks 2026 • Built in 24 hours • 500+ Participants • 2026

- Built a full-stack media bias detection platform in 24 hours, placing 3rd in Tier 2 among 500+ competitors.
- Analyzes articles, audio, and YouTube content for political bias, emotional manipulation, and inflammatory language using a custom "Drama Index" (1–100 scale) powered by RoBERTa emotion models and Gemini LLM.
- Features include real-time bias scoring, unbiased rephrasing suggestions, HTML diff highlighting, outlet bias lookup, and chunked YouTube audio transcription.
- Stack: Python, Flask, React, Gemini API, Hugging Face Transformers, Google Speech Recognition, yt-dlp.

Neural Network — House Price Predictor:

[Neural Network from Scratch](#) • [Paper In Progress](#) • [2026](#)

- A neural net that predicts residential property prices in Calgary, written end-to-end in NumPy — no Keras, no PyTorch — on the City's open assessment data.
- $R^2 = 0.986$ in log space, $R^2 = 0.87$ in dollar space, MAE ~\$66k, median APE ~6% on the held-out test set.
- Technical report in progress which covers all technical details.

LEADERSHIP & EXTRACURRICULARS

COO At usamoguide (January 2026 - Present):

[Remote](#)

- co-founded usaco.guide for math, a competition math platform with 50k+ site visits and 5k+ users.
- Developed full-stack in Next.js, including the MDX content pipeline and curriculum architecture.
- Actively publish comprehensive content modules to provide a structured roadmap in learning competition math.

Verve Consulting — Software Engineer

[Forbes-Featured Nonprofit](#) • [2026 – Present](#)

- Work with a team to develop digital and strategic solutions for a Forbes-featured nonprofit.
- Lead development and produce material such as grants, strategies, and blog posts.

CodeTheCure — Software Developer

[Cancer Research Startup](#) • [2026 – Present](#)

- A cancer research startup working with researchers from Harvard, Stanford, and MIT.
- Integrate AI/ML using PyTorch and APIs, work on UI/UX and functionality.

STEM Engagement Program — Mentor & Facilitator

[Westmount Charter School](#) • [2025 – Present](#)

- Mentored 100+ gifted middle school students in mathematics and problem-solving as part of a formal school outreach program.
- Designed and led sessions to make advanced mathematical thinking accessible and exciting for younger students.

Math Club — Incoming President

[Westmount Charter School](#)

- Elected as next president; plans include expanding competition math training and growing club membership.

Coding Club — Incoming President

[Westmount Charter School](#)

- Elected as next president; plans include expanding competitive programming training and growing club membership.

Reach for the Top — Team Member

[Westmount Charter School](#)

- Compete in academic trivia covering current events, history, science, literature, and pop culture.
- Team placed first in the province of Alberta.

SKILLS

Languages: Next.js, React, TypeScript, JavaScript, Python, C++, C, SQL

Frameworks/Technologies: Tailwind CSS, MDX, Flask, Supabase, Vercel, CI/CD, Git/GitHub,

Domains: Full-Stack Web Dev, Algorithms & Data Structures, Competitive Programming, Math Olympiad

Human Languages: English (Fluent), Hindi (Fluent), French (Semi-Proficient)